

The Perturbation Catalogue

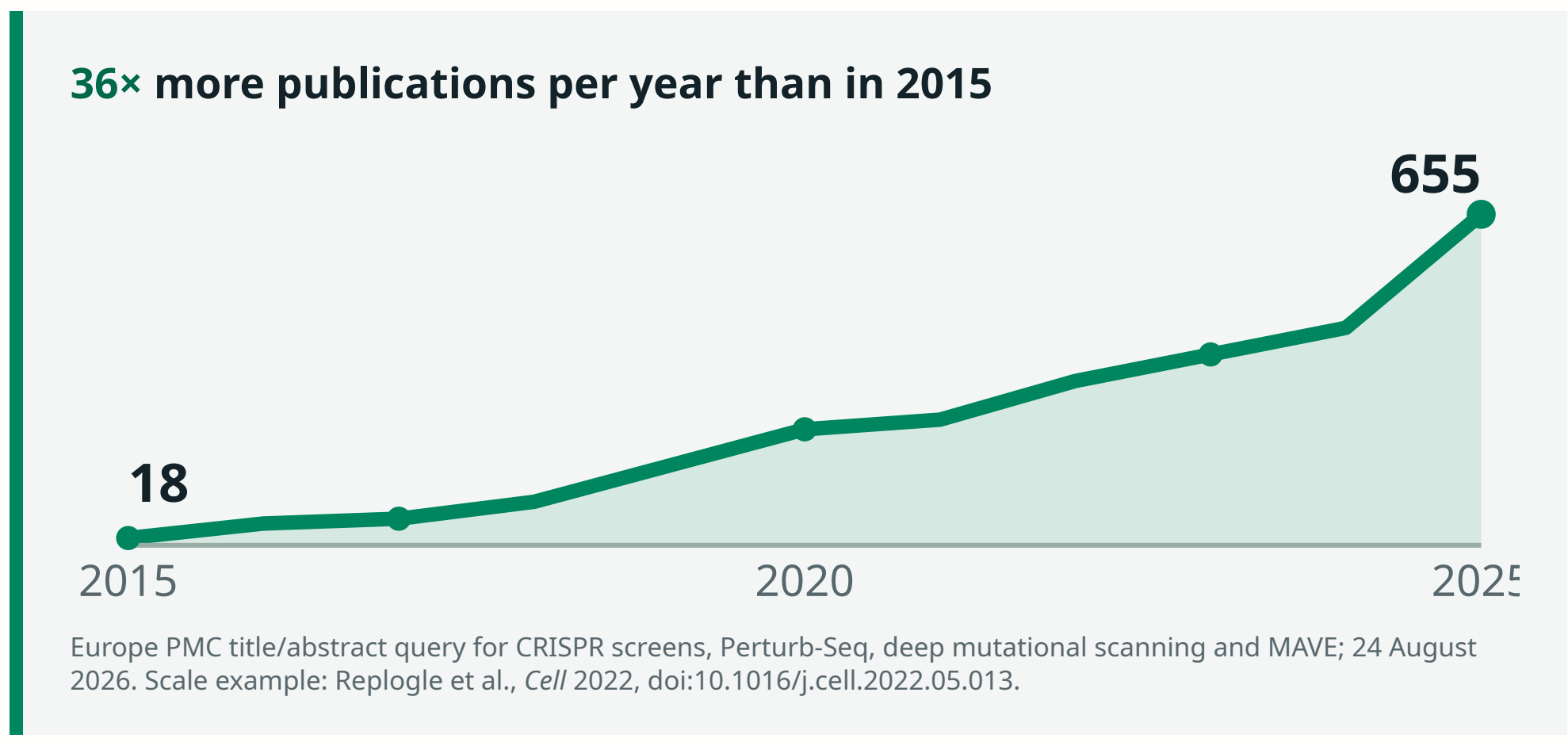
An Integrated Resource for Exploring the Impact of Genetic Perturbations

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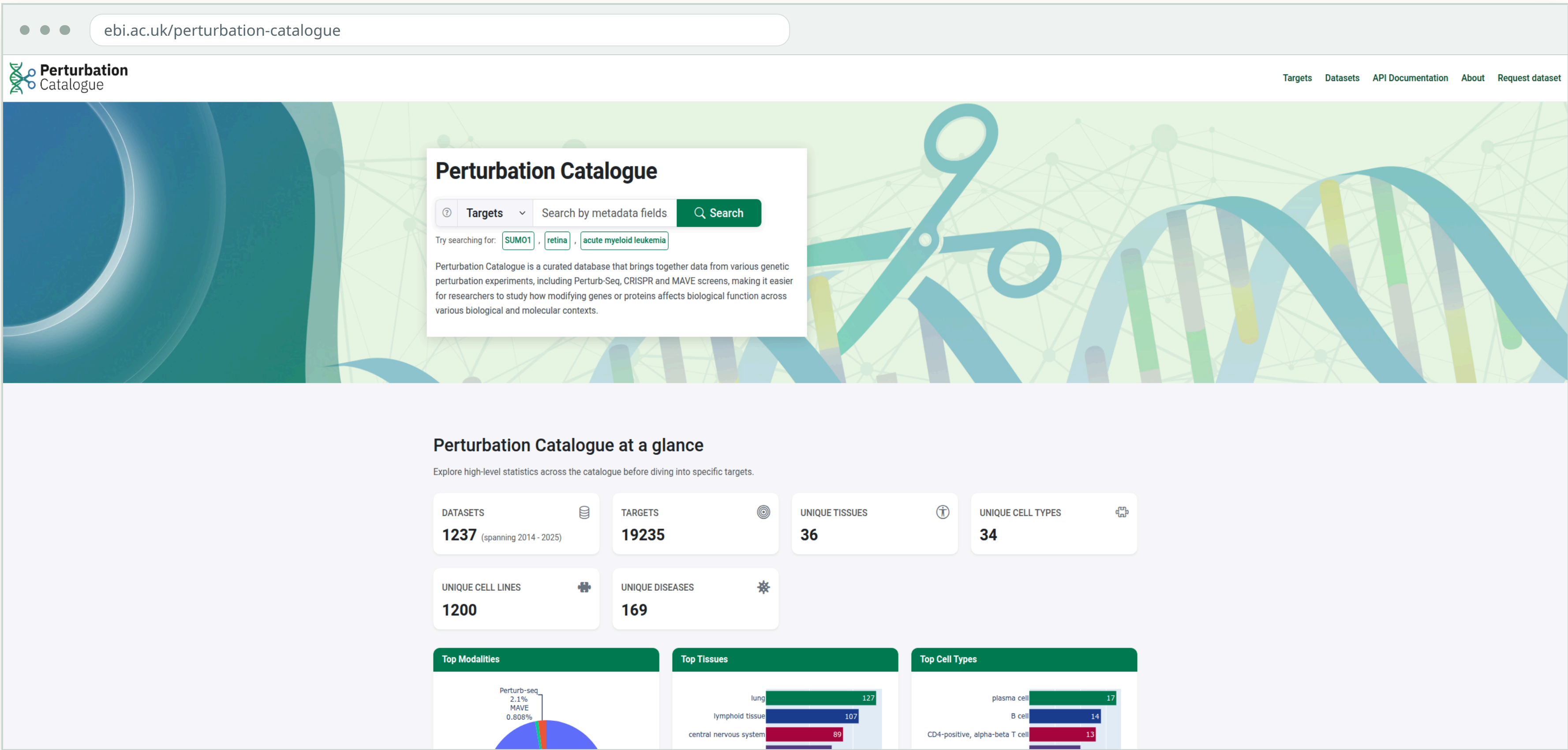
01 Perturbation evidence is multiplying faster than it can be organised

Screens now connect genetic changes to fitness, molecular function and whole-transcriptome responses at unprecedented scale. One genome-wide Perturb-Seq study profiled **more than 2.5 million cells**.

Different repositories / different identifiers / incomplete metadata / study-specific scores / incompatible processing



02 The Catalogue turns scattered studies into one place to search, interpret and retrieve



The production homepage: target search, current coverage and biological-context summaries.

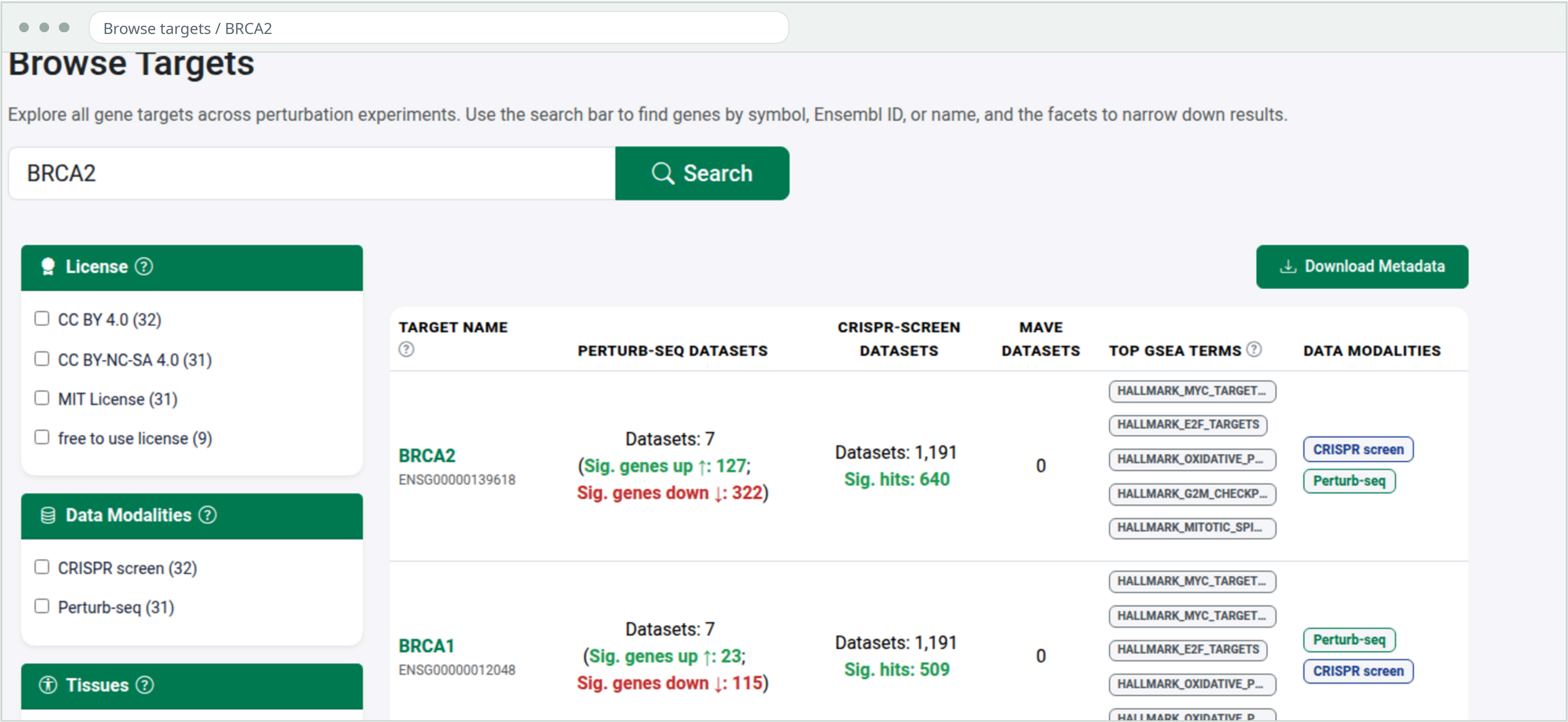
- One metadata contract**
Experimental design, perturbation, assay, model system, provenance and licence are validated against a shared schema and linked to ontologies.
- One stable target identity**
Ensembl gene IDs connect evidence across gene-level screens, variant measurements and transcriptomic responses.
- One access layer**
The portal and REST API expose searchable results, JSON metadata and complete CSV or Parquet datasets.

03 Today, 1,237 datasets connect three complementary modalities

1,237 datasets	19,235 human targets	36 tissues	34 cell types	1,200 cell lines	169 diseases
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- CRISPR screens 1,201 datasets**
Gene-level phenotype and fitness scores, significant hits and curated experimental context.
- Perturb-Seq 26 datasets**
Single-cell responses with differential expression and pathway enrichment; five datasets reprocessed from raw FASTQ.
- MAVE 10 datasets**
Variant-level functional measurements across protein sequence, searchable scores and effect heatmaps.

Production snapshot: 24 August 2026. All modalities use normalised Ensembl gene IDs.



One target view connects modality coverage, contributing datasets, significant effects and enriched pathways.

04 Next, integration becomes continuous—and reanalysis becomes broader

Curate faster, keep human judgement

AI-assisted extraction will draft structured metadata; curators will resolve ambiguity, validate ontology mappings and approve every release.

Outcome: richer metadata with a shorter path from publication to Catalogue.

Reanalyse more, measure quality

More kits and experimental designs, author-independent quality metrics and modality-appropriate reanalysis beyond the first Perturb-Seq datasets.

Outcome: results that are easier to compare—and easier to trust.

Release predictably, preserve provenance

Periodic, versioned data products will make additions and analytical changes explicit, reproducible and easier to adopt downstream.

Outcome: stable data products for both exploration and production workflows.

Bring us the next hard perturbation-data problem

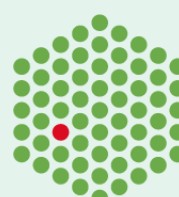
Suggest a dataset, challenge the schema or build the resource with us. We welcome scientific partnerships and funded multi-year collaborations around the modalities, analyses and functionality your programme needs.

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Explore the Catalogue



EMBL-EBI



Open Targets